We built a system that listens like a brain.
Not a neural network guessing patterns — a **glass-box cognitive architecture** that computes what happens inside your mind when you hear music. Every prediction, every surprise, every shiver — calculated in real time.

R³ The Ear
97 dimensions · 5.8ms

Breaks sound into 97 perceptual features every 5.8 milliseconds. Consonance, roughness, brightness, timbre, rhythm, harmony. Pure physics. No memory, no opinion. Frozen.

H³ Temporal Memory
8,600 active · 32 time scales

Watches how every feature evolves across 32 time windows — from 25ms (a single note) to 60 seconds (an entire section). Speed, acceleration, periodicity, trend, entropy. Frozen.

C³ The Cognitive Brain
1,006 dimensions · 96 models

96 computational models, 131 beliefs, 26 brain regions, 4 neurochemicals. Predicts what comes next, measures surprise when reality differs, updates beliefs, computes pleasure. Every frame.

MUSICAL INTELLIGENCE	TYPICAL MIR	NEURO RESEARCH
1,006D	~50D	~150D
	Spotify, Shazam	Academic labs

9 COGNITIVE FUNCTIONS — RUNNING IN PARALLEL

F1 **Sensory** Perceiving sound	F2 **Prediction** What comes next?	F3 **Attention** Filtering what matters
F4 **Memory** Have I heard this?	F5 **Emotion** How does this feel?	F6 **Reward** Pleasure & dopamine
F7 **Motor** Body responds	F8 **Learning** Adapting over time	F9 **Social** Shared experience

448	1,116	634	756	249
PEER-REVIEWED STUDIES	EMPIRICAL CLAIMS	EFFECT SIZES	PYTHON FILES	FRAMES PER SECOND

EVERY 5.8 MILLISECONDS

Predict → Observe → Surprise → Update → Reward

The brain predicts what sound comes next. When reality differs, **prediction error** fires. Beliefs update through **Bayesian inference**. Pleasure = surprise that resolves beautifully.

WHY IT MATTERS

Current music AI (Spotify, Magenta, Jukebox) uses **black-box neural networks** — billions of parameters, no understanding of *why* a song moves you.

We built a **glass box**: every parameter named, every decision traceable, every computation grounded in **published neuroscience**. When you get chills from a chord change, we can tell you exactly which brain region fired, which neurochemical shifted, and why.

- DA Dopamine — the thrill of surprise
- NE Norepinephrine — your curiosity engine
- OPI Opioids — the warmth of beauty
- 5HT Serotonin — patience & comfort

Manuscript target: Nature Neuroscience · 416 references · PRISMA 2020
26 brain regions (MNI152) · 131 beliefs · 96 models · 4 neurochemicals

"Music is not heard. It is thought."
— Amac Erdem, 2025